

Restaurant Success Analytics

Exploratory Analysis of Restaurant Trends and Customer Preferences Using Zomato Dataset

Submitted By

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Introduction

The restaurant industry has experienced significant growth over the past decade due to changing consumer preferences, increasing urbanization, and the rapid expansion of digital food delivery platforms. With customers having access to a wide variety of restaurants and cuisines, understanding customer behavior, restaurant performance, cuisine popularity, and pricing strategies has become essential for making informed business decisions.

This project analyzes the Zomato restaurant dataset to identify trends related to restaurant success, customer satisfaction, online delivery services, cuisine preferences, and pricing behavior. Using Exploratory Data Analysis (EDA), statistical techniques, data visualization, and feature engineering, the project aims to uncover meaningful insights and provide data-driven recommendations for improving restaurant performance and customer engagement.

Project Objectives

- Analyze restaurant distribution across different cities.
- Identify the most popular cuisine categories.
- Study restaurant ratings and customer satisfaction patterns.
- Examine the impact of online delivery services on restaurant performance.
- Analyze restaurant pricing behavior and cost distribution.
- Identify relationships between key business variables using correlation analysis.
- Create meaningful business features through feature engineering.
- Generate actionable business insights and recommendations for Zomato management.

Phase 1: Dataset Understanding

Dataset Overview

	Restaurant ID	Restaurant Name	Country Code	City	Address	Locality	Locality Verbose	Longitude	Latitude	Cuisines	...	Currency	Has Table booking	Has Online delivery	Is delivering now
0	6317637	Le Petit Souffle	162	Makati City	Third Floor, Century City Mall, Kalayaan Avenu...	Century City Mall, Poblacion, Makati City	Century City Mall, Poblacion, Makati City, Mak...	121.027535	14.565443	French, Japanese, Desserts	...	Botswana Pula(P)	Yes	No	No
1	6304287	Izakaya Kikufuji	162	Makati City	Little Tokyo, 2277 Chino Roces Avenue, Legaspi...	Little Tokyo, Legaspi Village, Makati City	Little Tokyo, Legaspi Village, Makati City, Ma...	121.014101	14.553708	Japanese	...	Botswana Pula(P)	Yes	No	No
2	6300002	Heat - Edsa Shangri-La	162	Mandaluyong City	Edsa Shangri-La, 1 Garden Way, Ortigas, Mandal...	Edsa Shangri-La, Ortigas, Mandaluyong City	Edsa Shangri-La, Ortigas, Mandaluyong City, Ma...	121.056831	14.581404	Seafood, Asian, Filipino, Indian	...	Botswana Pula(P)	Yes	No	No

3 rows × 21 columns

The dataset contains restaurant information including location, cuisine, ratings, votes, cost, delivery options, and customer feedback.

Dataset Characteristics

- Total Records: 9551
- Total Features: 21
- Numerical Features: 8
- Categorical Features: 13

Key Features

- Restaurant Name
- City
- Cuisines
- Aggregate Rating
- Votes
- Average Cost for Two
- Has Online Delivery
- Price Range

Observation

The dataset contains a diverse collection of restaurants from multiple cities and provides sufficient information for customer preference analysis.

Phase 2: Data Cleaning

Restaurant ID	0	<class 'pandas.core.frame.DataFrame'>
Restaurant Name	0	RangeIndex: 9551 entries, 0 to 9550
Country Code	0	Data columns (total 21 columns):
City	0	# Column Non-Null Count Dtype
Address	0	--- ---
Locality	0	0 Restaurant ID 9551 non-null int64
Locality Verbose	0	1 Restaurant Name 9551 non-null object
Longitude	0	2 Country Code 9551 non-null int64
Latitude	0	3 City 9551 non-null object
Cuisines	9	4 Address 9551 non-null object
Average Cost for two	0	5 Locality 9551 non-null object
Currency	0	6 Locality Verbose 9551 non-null object
Has Table booking	0	7 Longitude 9551 non-null float64
Has Online delivery	0	8 Latitude 9551 non-null float64
Is delivering now	0	9 Cuisines 9542 non-null object
Switch to order menu	0	10 Average Cost for two 9551 non-null int64
Price range	0	11 Currency 9551 non-null object
Aggregate rating	0	12 Has Table booking 9551 non-null object
Rating color	0	13 Has Online delivery 9551 non-null object
Rating text	0	14 Is delivering now 9551 non-null object
Votes	0	15 Switch to order menu 9551 non-null object
dtype: int64		16 Price range 9551 non-null int64
		17 Aggregate rating 9551 non-null float64
		18 Rating color 9551 non-null object
		19 Rating text 9551 non-null object
		20 Votes 9551 non-null int64
		dtypes: float64(3), int64(5), object(13)
		memory usage: 1.5+ MB

Data Cleaning Activities

- Handled missing values in Cuisines column.
- Verified categorical consistency.
- Checked duplicate records.
- Validated numerical features.

Findings

- Missing Values: 9 (Cuisines)
- Duplicate Records: 0

Observation

The dataset was found to be highly clean with minimal missing data and no duplicate records.

Phase 3: Descriptive Statistics

	Mean	Median	Mode	Std Dev	Variance
Average Cost for two	1199.210763	400.0	500.0	16121.183073	2.598925e+08
Aggregate rating	2.666370	3.2	0.0	1.516378	2.299401e+00
Votes	156.909748	31.0	0.0	430.169145	1.850455e+05
Price range	1.804837	2.0	1.0	0.905609	8.201274e-01

Variables Analyzed

- Average Cost for Two
- Aggregate Rating
- Votes
- Price Range

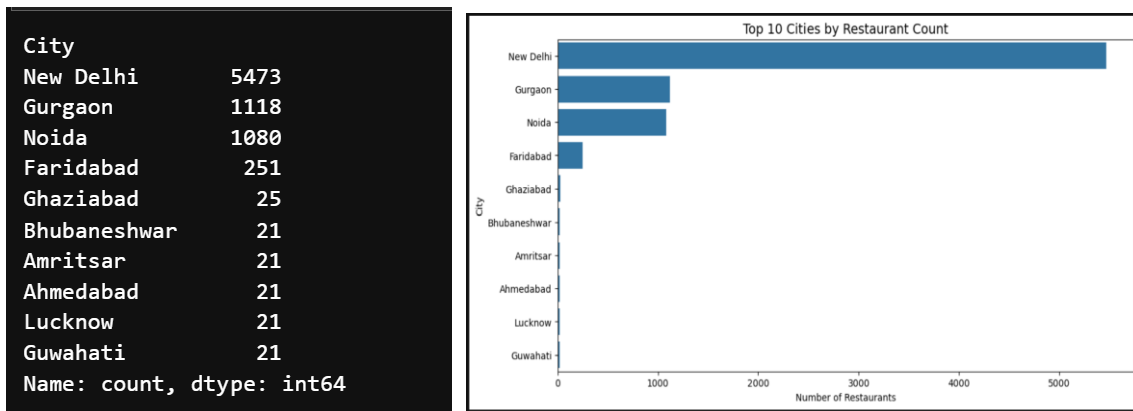
Key Findings

- Average restaurant cost is highly skewed.
- Customer votes vary significantly.
- Ratings show moderate variability.
- Most restaurants belong to affordable price categories.

Observation

Restaurant pricing and customer engagement exhibit significant variation across the dataset.

Phase 4: Restaurant Distribution Analysis



Findings

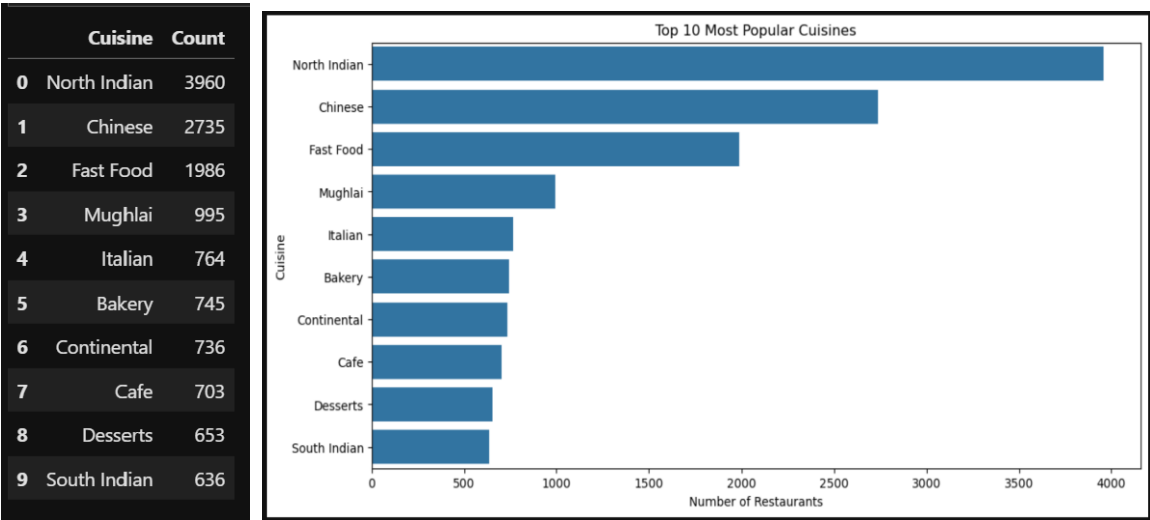
Top Cities:

- 1. New Delhi – 5473
- 2. Gurgaon – 1118
- 3. Noida – 1080
- 4. Faridabad – 251

Observation

New Delhi dominates the restaurant ecosystem and serves as the primary restaurant market within the dataset.

Phase 5: Cuisine Analysis



Top 10 Most Popular Cuisines

1. North Indian
2. Chinese
3. Fast Food
4. Mughlai
5. Italian
6. Bakery
7. Continental
8. Cafe
9. Desserts
10. South Indian

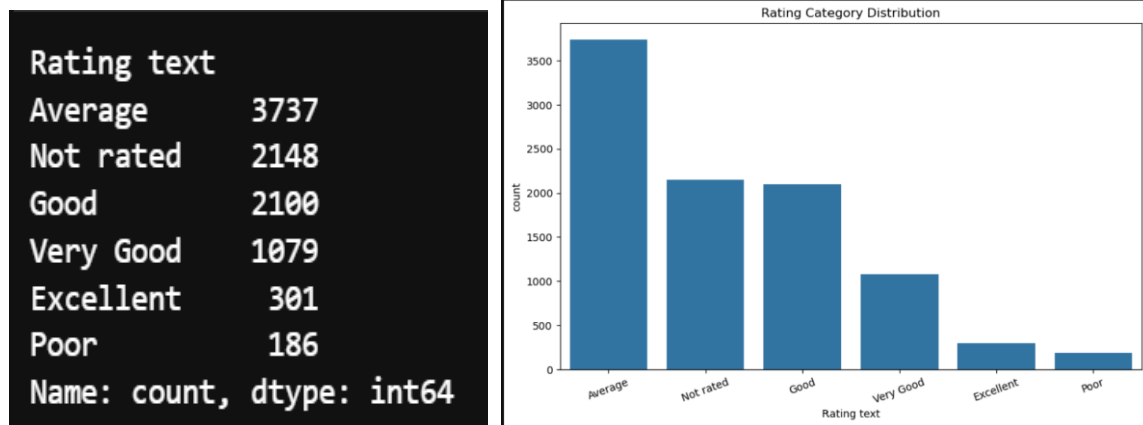
Key Finding

North Indian cuisine accounted for approximately 20% of cuisine occurrences.

Observation

Traditional Indian cuisine remains the most preferred food category among customers.

Phase 6: Rating Analysis



Rating Distribution

Most restaurants fall under:

- Average
- Good

categories.

Rating Category Counts

- Average: 3737
- Good: 2100
- Not Rated: 2148

Skewness

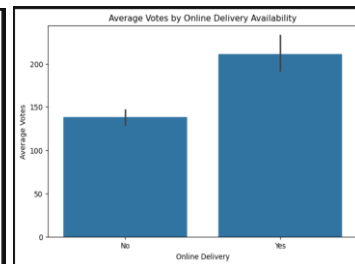
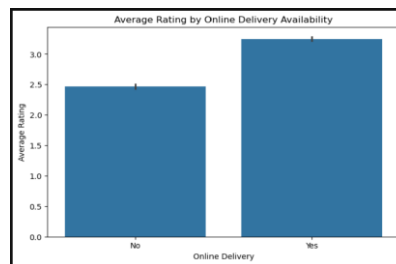
Rating Skewness = -0.95

Interpretation

Ratings are negatively skewed, indicating that most restaurants receive average-to-good customer feedback.

Phase 7: Online Delivery Analysis

```
Has Online delivery
No    7100
Yes   2451
Name: count, dtype: int64
```



Delivery Availability

- Delivery Restaurants: 2451
- Non-Delivery Restaurants: 7100

Average Ratings

- Delivery: 3.25
- Non-Delivery: 2.47

Average Votes

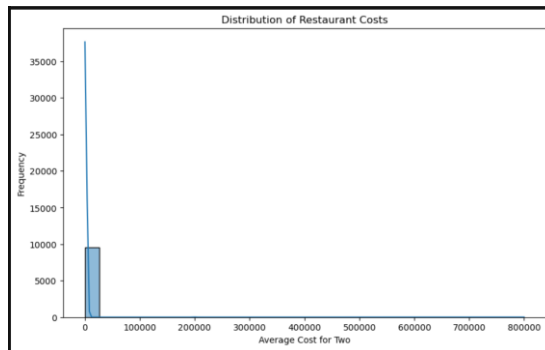
- Delivery: 211
- Non-Delivery: 138

Observation

Restaurants offering online delivery receive higher ratings and greater customer engagement.

Phase 8: Cost Analysis

	Restaurant Name	City	Average Cost for two	Aggregate rating	Votes
9283	Skye	Jakarta	800000	4.1	1498
9284	Satoo - Hotel Shangri-La	Jakarta	800000	4.6	873
9287	Sushi Masa	Jakarta	500000	4.9	605
9289	3 Wise Monkeys	Jakarta	450000	4.2	395
9294	Avec Moi Restaurant and Bar	Jakarta	350000	4.3	243
9288	Lucky Cat Coffee & Kitchen	Jakarta	300000	4.3	458
9295	Onokabe	Tangerang	300000	3.7	155
9278	Lemongrass	Bogor	250000	4.0	1159
9285	MONKS	Jakarta	250000	4.2	259
9279	Talaga Sampireun	Jakarta	200000	4.9	1662



Most Expensive Restaurants

Premium restaurants were primarily located in Jakarta and other international cities.

Price Range Distribution

- Price Range 1: 4444
- Price Range 2: 3113
- Price Range 3: 1408
- Price Range 4: 586

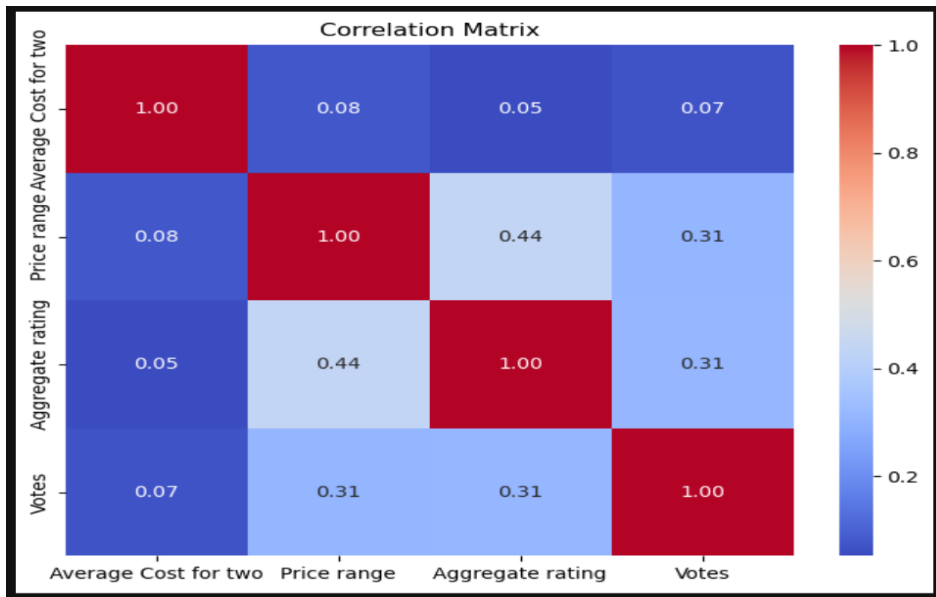
Observation

Affordable and moderately priced restaurants dominate the market.

Cost vs Rating

Restaurant ratings generally increase with price range.

Phase 9: Correlation Analysis



Correlation Findings

- Price Range ↔ Rating = 0.438
- Votes ↔ Rating = 0.314
- Price Range ↔ Votes = 0.309
- Cost ↔ Rating = 0.052

Observation

Price range and customer engagement show stronger relationships with ratings than restaurant cost.

Phase 10: Feature Engineering

Cost Category	
Budget	6074
Moderate	2893
Premium	552
Luxury	32
Name: count, dtype: int64	

Rating Category	
Average	3920
Good	3179
Poor	2151
Excellent	301
Name: count, dtype: int64	

New Features Created

Cost Category

- Budget
- Moderate
- Premium
- Luxury

Rating Category

- Poor
- Average
- Good
- Excellent

Cost Category Distribution

- Budget: 6074
- Moderate: 2893
- Premium: 552
- Luxury: 32

Rating Category Distribution

- Average: 3920
- Good: 3179
- Poor: 2151
- Excellent: 301

Observation

Luxury restaurants achieve the highest average ratings, while budget restaurants receive the lowest average ratings.

Business Insights

- New Delhi is the largest restaurant market and contributes the highest restaurant concentration.
 - North Indian cuisine is the most popular cuisine among customers.
 - Online delivery significantly improves ratings and customer engagement.
 - Restaurants receiving more votes generally achieve higher ratings.
 - Higher price categories tend to receive better ratings.
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Recommendations

- Encourage more restaurants to adopt online delivery services.
 - Promote high-demand cuisines such as North Indian, Chinese, and Fast Food.
 - Increase customer review participation through loyalty and feedback programs.
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Conclusion

- The Restaurant Success Analytics project successfully analyzed restaurant performance, customer preferences, pricing behavior, cuisine popularity, and delivery adoption trends.
- The analysis revealed that restaurant success depends on multiple factors including cuisine type, customer engagement, online delivery availability, and pricing segment.
- Online delivery emerged as one of the strongest drivers of restaurant performance, while cuisine popularity and customer feedback significantly influenced restaurant success.
- The findings provide valuable business insights for Zomato management and restaurant owners to improve customer satisfaction and operational performance.